

EXPERIMENT 12 : Earley Parser for Context-Free Grammar

Aim: To implement an Earley parser for a Context-Free Grammar using a simple Python program.

Program:

```
grammar = {
    "S": ["NP", "VP"],
    "NP": ["Det", "N"],
    "VP": ["V", "NP"],
    "Det": ["the", "a"],
    "N": ["student", "teacher", "book"],
    "V": ["reads", "likes"]
}

def earley_parse(words):
    n = len(words)
    chart = [[] for _ in range(n + 1)]
    chart[0].append(("S", ["NP", "VP"], 0, 0))
    for i in range(n + 1):
        changed = True
        while changed:
            changed = False
            for lhs, rhs, dot, origin in chart[i]:
                # Prediction
                if dot < len(rhs) and rhs[dot] in grammar:
                    next_symbol = rhs[dot]
                    for rule in grammar[next_symbol]:
                        item = (next_symbol, rule, 0, i)
                        if item not in chart[i]:
                            chart[i].append(item)
                            changed = True
```

```

# Completion
elif dot == len(rhs):
    for item in chart[origin]:
        lhs2, rhs2, dot2, origin2 = item
        if dot2 < len(rhs2) and rhs2[dot2] == lhs:
            new_item = (lhs2, rhs2, dot2 + 1, origin2)
            if new_item not in chart[i]:
                chart[i].append(new_item)
                changed = True

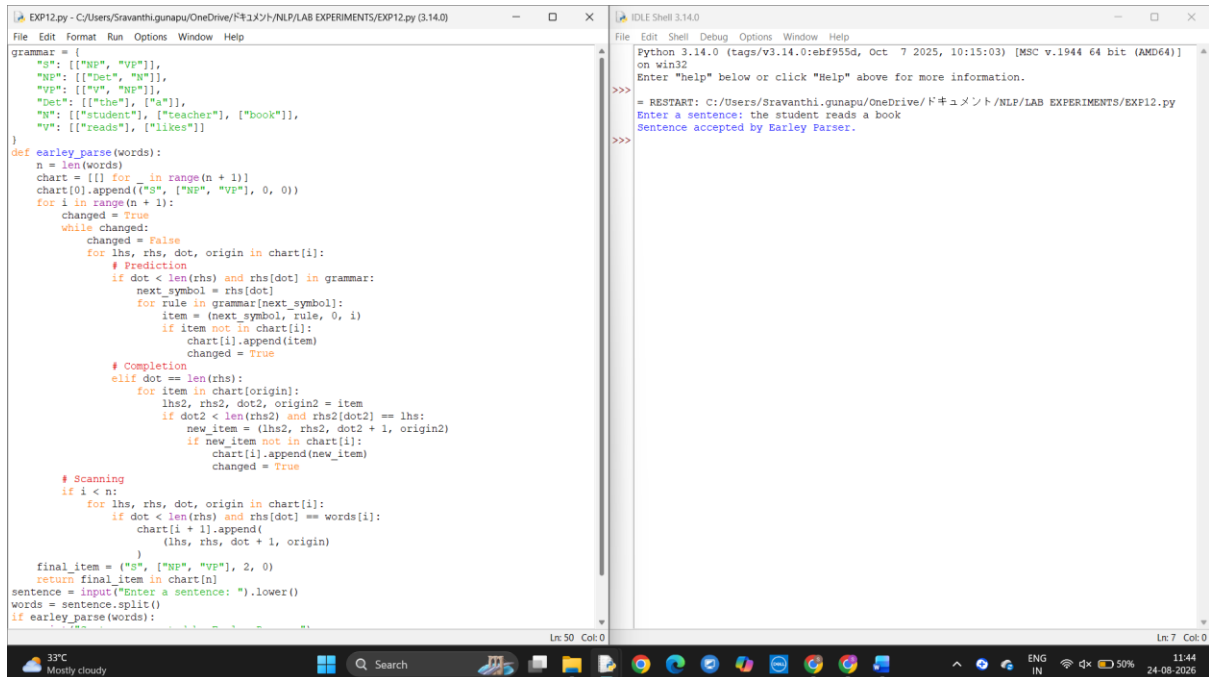
# Scanning
if i < n:
    for lhs, rhs, dot, origin in chart[i]:
        if dot < len(rhs) and rhs[dot] == words[i]:
            chart[i + 1].append(
                (lhs, rhs, dot + 1, origin)
            )

    final_item = ("S", ["NP", "VP"], 2, 0)
    return final_item in chart[n]

sentence = input("Enter a sentence: ").lower()
words = sentence.split()
if earley_parse(words):
    print("Sentence accepted by Earley Parser.")
else:
    print("Sentence rejected by Earley Parser.")

```

Output:



The screenshot displays a Python IDE with two windows. The left window, titled 'EXP12.py', contains the implementation of the Earley Parser. The right window, titled 'IDLE Shell 3.14.0', shows the execution output.

```
grammar = {
    "s": [{"NP", "VP"}],
    "NP": [{"Det", "N"}],
    "VP": [{"V", "NP"}],
    "Det": [{"the"}],
    "N": [{"student"}, {"teacher"}, {"book"}],
    "V": [{"reads"}, {"likes"}]
}

def earley_parse(words):
    n = len(words)
    chart = [[] for _ in range(n + 1)]
    chart[0].append(("s", ["NP", "VP"], 0, 0))
    for i in range(n + 1):
        changed = True
        while changed:
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            for lhs, rhs, dot, origin in chart[i]:
                # Prediction
                if dot < len(rhs) and rhs[dot] in grammar:
                    next_symbol = rhs[dot]
                    for rule in grammar[next_symbol]:
                        item = (next_symbol, rule, 0, i)
                        if item not in chart[i]:
                            chart[i].append(item)
                            changed = True
                # Completion
                elif dot == len(rhs):
                    for item in chart[origin]:
                        lhs2, rhs2, dot2, origin2 = item
                        if dot2 < len(rhs2) and rhs2[dot2] == lhs:
                            new_item = (lhs2, rhs2, dot2 + 1, origin2)
                            if new_item not in chart[i]:
                                chart[i].append(new_item)
                                changed = True
            # Scanning
            if i < n:
                for lhs, rhs, dot, origin in chart[i]:
                    if dot < len(rhs) and rhs[dot] == words[i]:
                        chart[i + 1].append(
                            (lhs, rhs, dot + 1, origin)
                        )
        final_item = ("s", ["NP", "VP"], 2, 0)
        return final_item in chart[n]
    sentence = input("Enter a sentence: ").lower()
    words = sentence.split()
    if earley_parse(words):
```

The right window shows the following output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP12.py
Enter a sentence: the student reads a book
Sentence accepted by Earley Parser.
>>>
```

The Windows taskbar at the bottom shows the date as 24-08-2025 and the time as 11:44.